

# Chapter 41 — Management Strategies for Vesicoureteral Reflux — Numbers, Gold Standards and Traps

## IF YOU READ NOTHING ELSE

- **VUR (vesicoureteral reflux) with UTI (urinary tract infection)** about 30%; other indications about 17%.
- **Gold standard diagnosis:** VCUG (voiding cystourethrogram) or direct RNC (radionuclide cystography); both need a catheter.
- **Gold standard parenchyma:**  $^{99m}\text{Tc}$ -DMSA (dimercaptosuccinic acid); scar sensitivity 98%, specificity 92%.
- **Sterile VUR does not scar** (Ransley-Risdon pig model).
- **Resolution:** grade I–II most; III about 50%; IV–V few.
- **RIVUR:** CAP (continuous antibiotic prophylaxis) halves recurrent UTI (23.6% → 12.9%); **scarring unchanged**.
- **Swedish Reflux Trial:** girls on surveillance 57% febrile UTI vs 19% CAP.
- **BBD** (bladder and bowel dysfunction) → recurrent UTI about 45% vs 15%; treat BBD first.
- **Breakthrough febrile UTI on CAP** → correct VUR.
- **Scars, not VUR**, drive pregnancy morbidity and hypertension.

## EMBRYOLOGY AND FUNCTIONAL ANATOMY

- **Ureteric bud:** weeks 4–6, from the mesonephric duct.
- **Mackie-Stephens:** early bud → high lateral orifice → VUR; late bud → ectopic, obstructed.
- **Paquin:** tunnel:diameter **5:1 normal vs 1.4:1 refluxing**.
- **Waldeyer's sheath:** outer ureteric + outer detrusor muscle → deep trigone.

## DEMOGRAPHICS

- **Infant boys predominate** (up to 3:1 at 0–6 months) → 1:1 by 2 years; overall girls about 2×.
- **Uncircumcised infant boys:** 10–12× UTI.
- **Circumcision NNT** (number needed to treat): 111 general vs 4 with VUR; topical steroid = circumcision.
- **VUR with UTI by age:** <2 y 38.6% · 2–6 y 26.6% · 7–11 y 19.7% · 12–21 y 7.6%.
- **Black children:** about 3× less; no grade IV–V under 6; equal after 10.
- **Prenatal hydronephrosis 1–5%;** VUR in 10–40% of screened infants.
- **Newborn VCUG only for:** high-grade or bilateral hydronephrosis, cortical change, dilated ureter, ureterocele, duplex, bladder anomaly.

## INHERITANCE AND GENETICS

- **Siblings about 30%** (to 50%); higher <3 years; identical twins near-complete concordance.
- **AUA** (American Urological Association): **sibling VCUG if** cortical abnormality, size asymmetry, UTI, or hydronephrosis grade 3–4.
- **No mass sibling screening.**

## GRADING

- **International classification (1981):** I ureter · II pelvis, not dilated · III mild dilatation · IV tortuosity · V gross, papillae lost.
- **RNC grades:** 1 = I; 2 = II–III; 3 = IV–V.
- **UDR** (ureteral diameter ratio) = max ureter in false pelvis ÷ L1–L3 distance; predicts resolution and BUTI (breakthrough UTI).
- **Interobserver agreement** poor in grades II–IV.

## AETIOLOGY

- **Spina bifida:** VUR up to 30% → early CIC (clean intermittent catheterisation) + anticholinergic.
- **Normal voiding about 100 cm H<sub>2</sub>O** without VUR → structural UVJ (ureterovesical junction) failure is key.
- **Detrusor overactivity** = commonest urodynamic finding with VUR.
- **Reflux during filling** = poor prognosis for resolution.

## DIAGNOSIS AND LOWER TRACT

- **VCUG filling height** ≤70 cm H<sub>2</sub>O.
- **RNC dose:** about 1% of continuous-fluoroscopy VCUG; misses grade I; best for follow-up.
- **Indirect RNC:** no catheter; needs toilet training.
- **Cyclic VCUG** → 12–20% more VUR detected.
- **VCUG timing:** once urine sterile and child improved (classically ≥1 week).
- **Contrast-enhanced urosonography:** no radiation; ≈ VCUG.
- **Routine cystoscopy:** no role.
- **PIC** (positional instillation of contrast): recurrent febrile UTI + negative VCUG; contrast from 1 m under anaesthesia; not first line.
- **Invasive urodynamics:** refractory cases only; uroflow + EMG (electromyography) non-invasive.
- **VUR + UVJ obstruction signs** → relative contraindication to endoscopic injection.

## UPPER TRACT AND CORTICAL DEFECTS

- **Ultrasound cannot diagnose VUR;** normal in about 25% of refluxing infants.
- **VUR about 3× cortical defects** after infection; VUR present in 30% of pyelonephritis.
- **Congenital damage:** grades IV–V, male infants, no UTI; grades I–III normal.
- **Greatest scarring risk:** first year; <4 years after a single UTI.
- **Big bang** (Ransley-Risdon): most scarring after the first pyelonephritis.
- **Compound papillae** at the poles → intrarenal reflux; neonate at 2 mm Hg.
- **Hypertension** in 34–38% with scars; reflux nephropathy = commonest secondary cause in children.
- **VUR correction does not cure hypertension;** segmental resection or nephrectomy may.
- **USRDS** (US Renal Data System): reflux nephropathy RRT (renal replacement therapy) 1.9 → 0.6 per million (1996–2014).

## GUIDELINES

- **AAP** (American Academy of Pediatrics) **2011**, febrile UTI 2–24 months: ultrasound first; VCUG after a **second** febrile UTI or abnormal ultrasound.
- **AAP culture threshold:** 50,000 CFU/mL (colony-forming units) + abnormal urinalysis; oral = parenteral.
- **NICE** (National Institute for Health and Care Excellence): >3 years → no routine VCUG; 6 months–3 years response within 48 h → no VCUG.
- **TDA** (top-down approach): DMSA first; VCUG only if abnormal.

- **Normal DMSA** → <1% high-grade VUR; abnormal in about 70% of febrile UTIs.

## ASSOCIATED ANOMALIES

- **UPJO** (ureteropelvic junction obstruction) in VUR 0.75–3.6%; high grade 5× more.
- **VUR + UPJO** → **fix obstruction first**.
- **Duplex**: VUR = commonest anomaly, lower pole (Weigert-Meyer).
- **Paraureteral diverticulum** → resolves like primary; **ureter into diverticulum** → won't resolve.
- **MCDK** (multicystic dysplastic kidney): contralateral VUR about 25%.
- **Renal agenesis**: contralateral VUR 24–28%.
- **Megacystis-megaureter**: non-obstructive; normal urethra; reimplant; **never operate on the outlet**.
- **Obstructive refluxing megaureter**: dilute ureteric contrast; injection may obstruct.

## FOLLOW-UP, ADULTHOOD AND PREGNANCY

- **Dx/HA** (dextranomer/hyaluronic acid) **mound calcifies in >50%** → mimics stone.
- **VUR in pregnancy**: pyelonephritis about 33% vs <5%.
- **Scars** → hypertension about 3×, pre-eclampsia about 8×.
- **Offspring** of affected mothers: VUR about 43%.

## NATURAL HISTORY

- **Grade II 63–85%**; grade III about 50%, about 90% of these within 4 years.
- **Observation period: 5 years**; age at presentation > grade as a predictor.

## LANDMARK STUDIES

- **IRSC** (International Reflux Study in Children): high grade, CAP vs surgery → UTI 38% both; scars equal.
- **IRSC 10 years**: equal growth and renal function.
- **Birmingham**: scars equal, all within 2 years.
- **Swedish**: grade III–IV, 1–2 y: girls 19% CAP, 23% endoscopic, 57% surveillance; boys no difference.
- **RIVUR**: 607, grade I–IV, TMP-SMX (trimethoprim-sulfamethoxazole) vs placebo; NNT high risk 5 vs low risk 18.

- **PRIVENT**: absolute UTI reduction 6% (19% → 13%).

## CURRENT MEDICAL MANAGEMENT

- **Open correction >95%** success.
- **CAP**: little benefit grade ≤II without BBD; benefit grade ≥III in girls.
- **15%** have febrile UTI within 2 years; 15% of these scar.
- **Off CAP**: 8–10% develop febrile UTI.
- **<2 months**: trimethoprim or amoxicillin; **no sulfonamide** (jaundice).
- **Nitrofurantoin**: interstitial lung disease (rare, full dose).
- **Dose at night**.
- **Smellie BUTI**: sensitive organism → non-adherence or low dose; resistant → high residual.
- **Hidas 2-year BUTI on CAP**: low 8.6%, intermediate 26%, high 62.5%.
- **High risk**: female, <1 y, white, bilateral, grade III–V, BBD, scars, multiple febrile UTIs.

## CLASSIC TRAPS

- **Trap**: VUR causes UTI — **Actually**: it accelerates pyelonephritis only.
- **Trap**: every renal defect with VUR is a scar — **Actually**: high-grade male infants have congenital dysplasia.
- **Trap**: normal ultrasound excludes VUR — **Actually**: ultrasound cannot diagnose VUR.
- **Trap**: CAP prevents scarring (RIVUR) — **Actually**: prevents UTI only.
- **Trap**: correcting VUR cures hypertension — **Actually**: parenchymal; correction does not.
- **Trap**: RNC detects all grades — **Actually**: misses grade I.
- **Trap**: fix VUR first with coexisting UPJO — **Actually**: relieve obstruction first.
- **Trap**: megacystis-megaureter = valves — **Actually**: non-obstructive; normal urethra on VCUG.
- **Trap**: TMP-SMX in a 1-month-old — **Actually**: sulfonamide jaundice; use trimethoprim or amoxicillin.
- **Trap**: one negative fill-void cycle excludes VUR — **Actually**: cyclic study finds 12–20% more.